U.S Traffic Fatality Explorer

A nationwide visualization of fatal injuries suffered in motor vehicles traffic crashes

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Research Background: Every year, nearly 40,000 people die on U.S. roads, many of which are preventable through better enforcement, laws, and road design [15]. From passengers to pedestrians, people are constantly at risk for a variety of reasons, including location-related reasons such as weather, lighting or road conditions, and safety-related conditions such as speeding, alcohol and drug involvement, and much more. With a variety of different factors responsible for crashes, we believe it's important to be able to be informed on key factors to be able to make more informed decisions on road-related things to ensure a safer traffic environment for us all.

This proposal will utilize the NHTSA Fatality Analysis Reporting System (FARS), which covers every crash in where someone died within 30 days of a motor vehicle related incident in the United States. This dataset includes crash details, people and vehicle details, and other crash-related conditions such as speeding or alcohol involvement.

Design Requirements: Our goal is to reduce the amount of traffic-related incidents through better information and understanding of that information. To do this, we hope to help users identify when, where, and under what conditions fatal crashes occur.
Overall, there are a lot of different people involved in road accidents, from government officials to make a decision on road safety, police officers to enforce laws, city planners for infrastructure, and drivers who are actively on the road. Each of these groups are interested in different things, but we hope to make a centralized system that people can use to justify decisions and learn potential ways to contribute to a safer traffic environment.

For each of these four groups, (1) Drivers, (2) Government officials, (3) Police Officers, and (4) City Planners, we intend on interacting with them in the following ways to better understand the design requirements and what they would like to see in the visualizations.

1. **Drivers:** We plan to conduct a campus-wide survey to better understand what road-safety issues drivers are most concerned or uncertain about. Questions may focus on topics such as dangerous driving conditions, speeding, impaired driving, pedestrian safety, and locations or times they perceive as higher risk. These responses can help determine which information would be most useful to present to everyday road users. The survey can be found here, and has been currently used on a small sample set of people.
2. **Government Officials:** We will examine NHTSA and FHWA decision-making and safety-performance guidance to understand what information government officials use when evaluating traffic-safety problems and allocating resources. These sources emphasize measures such as fatalities, high-risk locations, affected populations, contributing crash factors, and changes following safety interventions.

3. **Police Officers:** We will analyze existing law-enforcement crash-analysis tools, such as the Michigan Traffic Crash Analysis Tool, along with related research on law-enforcement data dashboards. These systems provide insight into the types of information officers use, including crash locations, time patterns, speeding, alcohol involvement, crash types, and trends that may help guide enforcement decisions. We intend on reaching out to the College Station police department to conduct a survey on what things they look at to make a decision.
4. **City Planners:** We will review transportation-planning studies, practitioner surveys, and FHWA safety-planning guidance to identify the information planners use when making infrastructure decisions. Important factors include crash hotspots, roadway characteristics, pedestrian and cyclist safety, environmental conditions, and changes in crash patterns before and after roadway improvements. Currently, we are trying to find civil engineers with experience in traffic planning and conduct an interview to better understand the requirements.

From our current understanding of the information we have researched, from what our interest groups look at, we have came up with these core design requirements and design tasks:

Design Requirements	Design Tasks
Crash data should be interactive, geographic, filterable, and easy to compare [2][3][6][7][8].	Filter crash data by location, time, and crash factors; explore crashes on a map; compare locations side-by-side.
Fatal-crash patterns should be easy to identify across locations, time periods, and conditions[1][4][5][8].	Compare fatal-crash patterns across locations; analyze crashes by hour, day, month, and year; compare crashes under different conditions.
Users should be able to quickly identify high-risk areas and recurring crash hotspots[1][4][5][8].	Locate fatal-crash hotspots; rank areas by crash frequency; identify recurring high-risk locations over time.
The visualization should make it easy to compare different cities, counties, roads, or time periods[2][3][4][8].	Compare cities, counties, or roads; compare two or more time periods; identify differences in crash patterns between locations.
Contributing factors such as speeding, alcohol, weather, lighting, and road conditions should be clearly visible [1][4][10].	Analyze contributing factors; determine which factors are most common in a selected area; compare factor prevalence across locations or time periods.
Users should be able to filter multiple crash factors together to investigate specific safety concerns [2][3][6][7][8].	Combine multiple filters; investigate scenarios such as alcohol-related nighttime crashes or pedestrian crashes in poor weather; narrow results to specific conditions.
Trends over time should be easy to understand so users can see whether safety conditions are improving or worsening [4][5][8][9]	Analyze yearly crash trends; identify increases or decreases over time; compare trends across locations or crash types.
Information about pedestrians, motorcyclists, drivers, and other vulnerable road users should be easy to explore [1][4][5][10]	Compare road-user groups; analyze pedestrian and motorcycle fatalities; examine how risk differs across locations, time periods, or conditions.

Users should be able to query particular crash types and locations, and they can use dashboards to examine trends over time [3][8].	Query specific crash types; select a location and inspect related crashes; analyze trends for that selected crash type or location.
Important patterns should be clear and actionable for enforcement, infrastructure, and policy decisions. [4][5][6][7][9]	Identify high-risk locations; summarize major contributing factors; compare locations and trends; highlight patterns that may support resource-allocation decisions.
The system should support filtering before and after certain dates to check for changes in crashes [4][5][9].	Compare crashes before and after a selected date; examine changes following a law, enforcement effort, or infrastructure project; analyze whether trends shifted after the selected date.
Visualizations should be simple enough for general drivers while still useful for planners, police, and government officials [6][7][8][9]	Provide high-level summaries for casual users; allow deeper filtering and comparison for advanced users; support both overview and detailed exploration.

Related Work

1. **NHTSA FARS Query Tool:** It is an official federal database, which has nationwide fatal crash records from 1975-2024, but it is mostly based on static queries and raw data downloads, not dynamic visual cross filtering.
2. **NYC Vision Zero View:**It is a web based public safety map that displays citywide traffic deaths to allow for street redesigns but it doesn't have organized multi view filtering for factors such as driver impairment, speeding and road conditions.
3. **Traffic Origins by Anwar et al. (Research Paper):** This paper introduces the development of a visual analytics system to monitor spatio-temporal road incidents and traffic congestion, for real time clearance purposes rather than long term infrastructure planning or fatal risk analysis.
4. **Michigan Traffic Crash Analysis Tool:** This tool is a crash analysis tool specifically used by police officers in Michigan and features a system they actually use to support law enforcement officers with real-time data.

Novelty

Our tool makes exploring and analyzing national crash data accessible and intuitive by linking together interactive maps, temporal trends, and safety factors. Our approach goes beyond existing tools by providing a multi-layered perspective that drills down into relevant factors. In contrast to existing tools that display tabular data or only focus on one city, our tool links data such as clicking on a map location or risk factor updates related bar charts. We also enable comparison of fatality risk across counties or states side-by-side. We further allow for evaluating whether the number of crashes has declined following a particular law enforcement or roadway change on a given date. This tool offers both the layman an overview of risky locations while also enabling traffic planners and police to delve into the details to understand the contributing factors in order to improve safety.

Development Plans

The U.S. Traffic Fatality Explorer will be developed collaboratively by four team members using the 2024 Fatality Analysis Reporting System (FARS) dataset. Development will involve data preparation, visualization, dashboard integration, and evaluation.

Task Assignments

- **Varshini Ranga Swamy (Data Engineering):** Merge FARS crash, vehicle, and person files, decode variables, and create indicators for speeding, alcohol involvement, pedestrian and motorcycle crashes, lighting, and road types.
- **Lawrence Wong (Spatial Analysis):** Develop interactive crash hotspot maps by state and county, calculate fatal crash rates per 100,000 people, and enable comparisons between high-risk areas.
- **Utkarsh Bhagat (Temporal and Factor Analysis):** Develop visualizations of crash patterns by hour, day, and month, along with contributing factors such as speeding, alcohol, weather, lighting, and roadway conditions.
- **Vasav Srivastava (Integration and Vulnerable Road Users):** Integrate visualizations into an interactive dashboard with linked filters and dedicated views for pedestrian and motorcycle fatalities.

All members will contribute to dashboard evaluation, testing, and final project deliverables.

Development Timeline

Timeline	Planned Activities
Oct 5 to 11	Prepare FARS 2024 data, merge files, decode variables, and create crash-factor indicators.
Oct 12 to 25	Develop crash hotspot maps, temporal visualizations, and contributing-factor charts.
Oct 26 to Nov 8	Integrate linked filtering and develop pedestrian and motorcycle views.
Nov 9 to 15	Add high-risk area comparisons and roadway characteristic analysis.
Nov 16 to 22	Evaluate decision-making scenarios, gather feedback, and refine the dashboard.
Nov 23 to early Dec	Complete the dashboard, final report, demonstration, and presentation.

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